

2.1.4 Nucleic acids

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the structure of a nucleotide as the monomer from which nucleic acids are made	To include the differences between RNA and DNA nucleotides, the identification of the purines and pyrimidines, the type of pentose sugar and the formation of phosphodiester bonds (the sugar phosphate 'backbone').
(b) the structure of adenosine di-phosphate (ADP) and adenosine tri-phosphate (ATP) as phosphorylated nucleotides	
(c) (i) the structure of the DNA molecule, including a review of the evidence for complementary base pairing (Chargaff's rules)	
(ii) practical investigation into the purification of DNA by precipitation	M0.3
(d) semi-conservative DNA replication	HSW3, HSW7
(e) the nature of the genetic code	To include the roles of the enzymes helicase and DNA polymerase, the importance of replication in conserving genetic information with accuracy and the occurrence of random, spontaneous mutations.
(f) the structure of RNA (ribonucleic acid) and how it differs from that of DNA	To include reference to the triplet, non-overlapping, degenerate and universal nature of the code and how a gene determines the structure of proteins including enzymes by ordering the sequence of amino acids in a polypeptide.
(g) transcription and translation of genes resulting in the synthesis of polypeptides.	To include the role of RNA polymerase, messenger (m)RNA, transfer (t)RNA and ribosomal (r)RNA (details of post transcriptional modification are not required).